🛡️ Module 08: Windows Exploitation and Privilege Escalation

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# Exercise 08: Abusing UAC High Integrity Level
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Scenario

Windows User Account Control (UAC) enforces process isolation by assigning one of four integrity levels (IL) to token objects: **Low** (restricted browser sandbox), **Medium** (standard users and administrators' filtered tokens), **High** (administrators' elevated tokens), and **System** (OS services). When an administrative user logs in, Windows issues two tokens: a **Filtered Token** (Medium IL) for daily operations and an **Elevated Token** (High IL) reserved for administrative actions.

Certain native administrative applications utilize **Auto-Elevation** to launch silently without presenting a UAC consent dialog. In this exercise, we analyze token properties and mitigation policies (such as Data Execution Prevention / DEP enforcement status: **False** for elevated shells) using **Process Hacker 2** on the **Sales Department** machine. We exploit Auto-Elevated utilities (`msconfig.exe` and `azman.msc` via `mmc.exe`) to spawn administrative High-Integrity command prompts (`cmd.exe`) without triggering a UAC prompt.

Objective

The objective of this lab is to help students learn how to:
- Differentiate between UAC Integrity Levels (Low, Medium, High, System) and token types (Filtered vs. Elevated)
- Analyze process tokens, `SeDebugPrivilege` availability, and mitigation policies (DEP enforcement: **False**) using Process Hacker 2
- Identify default UAC notification levels and Auto-Elevation mechanics
- Abuse `msconfig.exe` (System Configuration) to launch an elevated Command Prompt (**High Mandatory Level**)
- Abuse `azman.msc` (Authorization Manager) via MMC Help file view-source options to spawn High-Integrity Command Prompts from Notepad

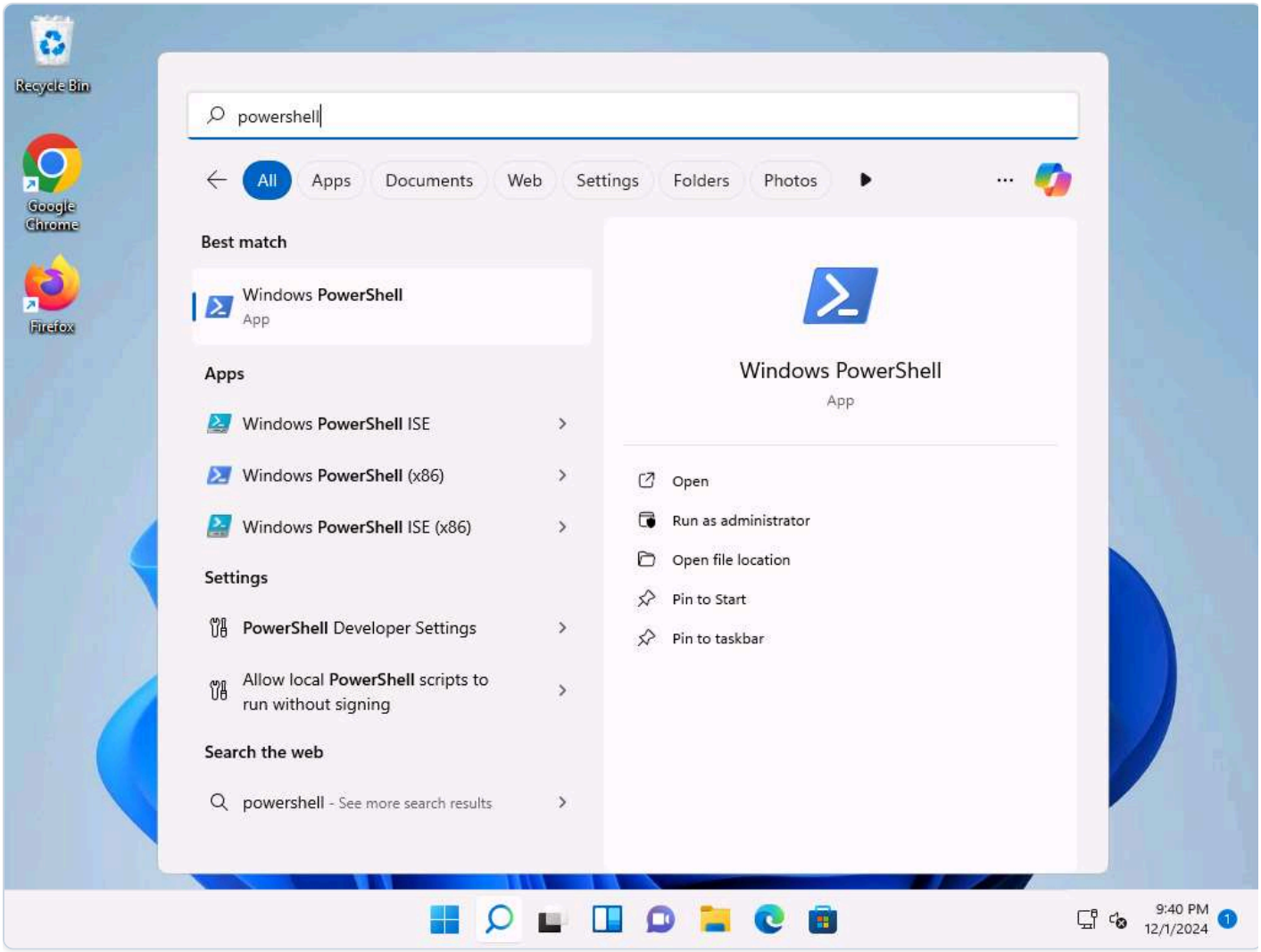
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Task 1: Analyze UAC Token Integrity Levels, Mitigation Policies, and Auto-Elevate Mechanics

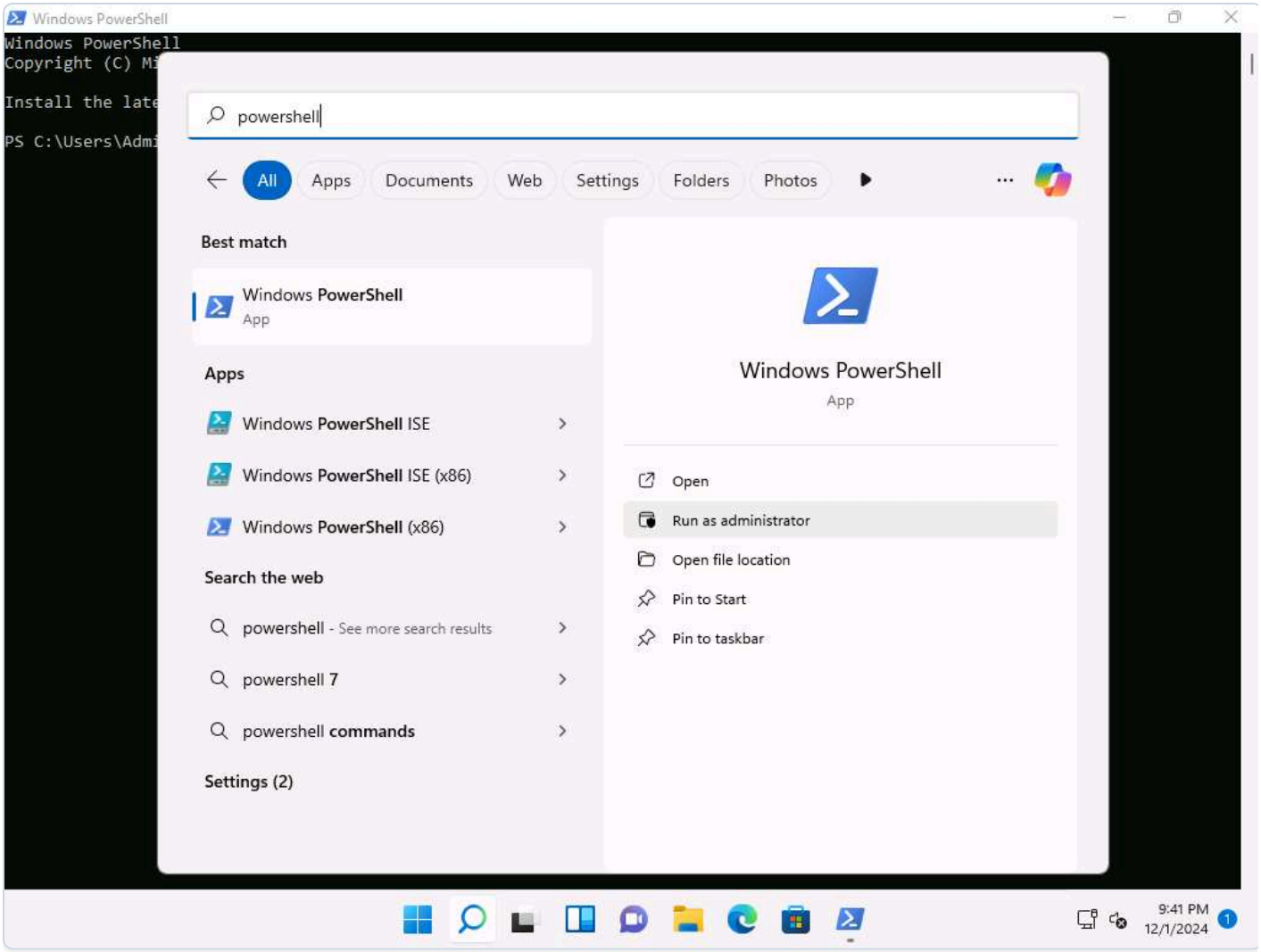
In this task, we launch standard and elevated PowerShell sessions on the Sales Department machine, compare token integrity levels and DEP status in Process Hacker 2, and demonstrate UAC auto-elevation bypasses via `msconfig.exe` and `azman.msc` .

Lab Tasks

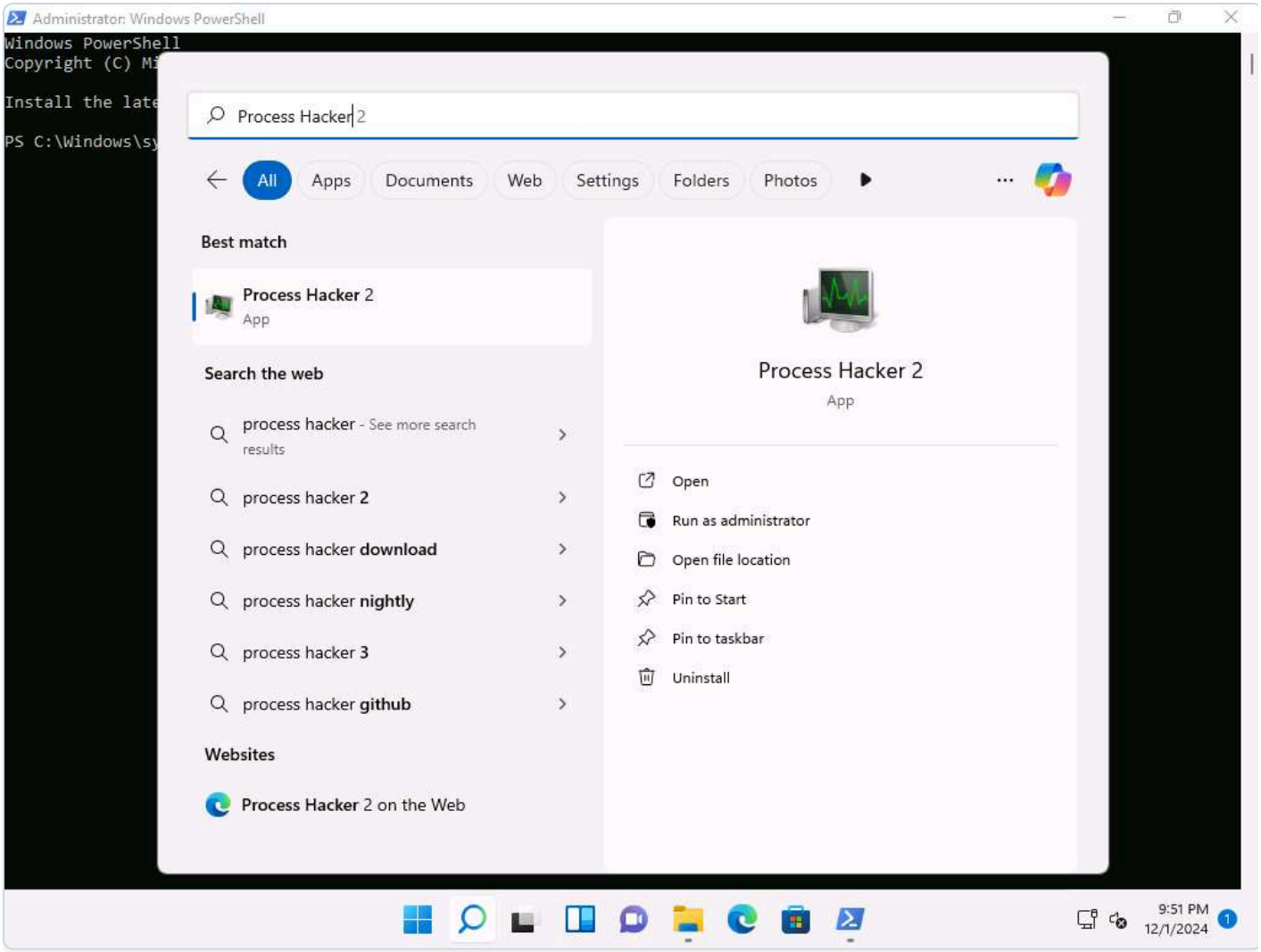
1. Switch to **Sales Department** machine and log in (**Admin** / **test@123**).
2. Open a standard (Medium IL) **Windows PowerShell** window.



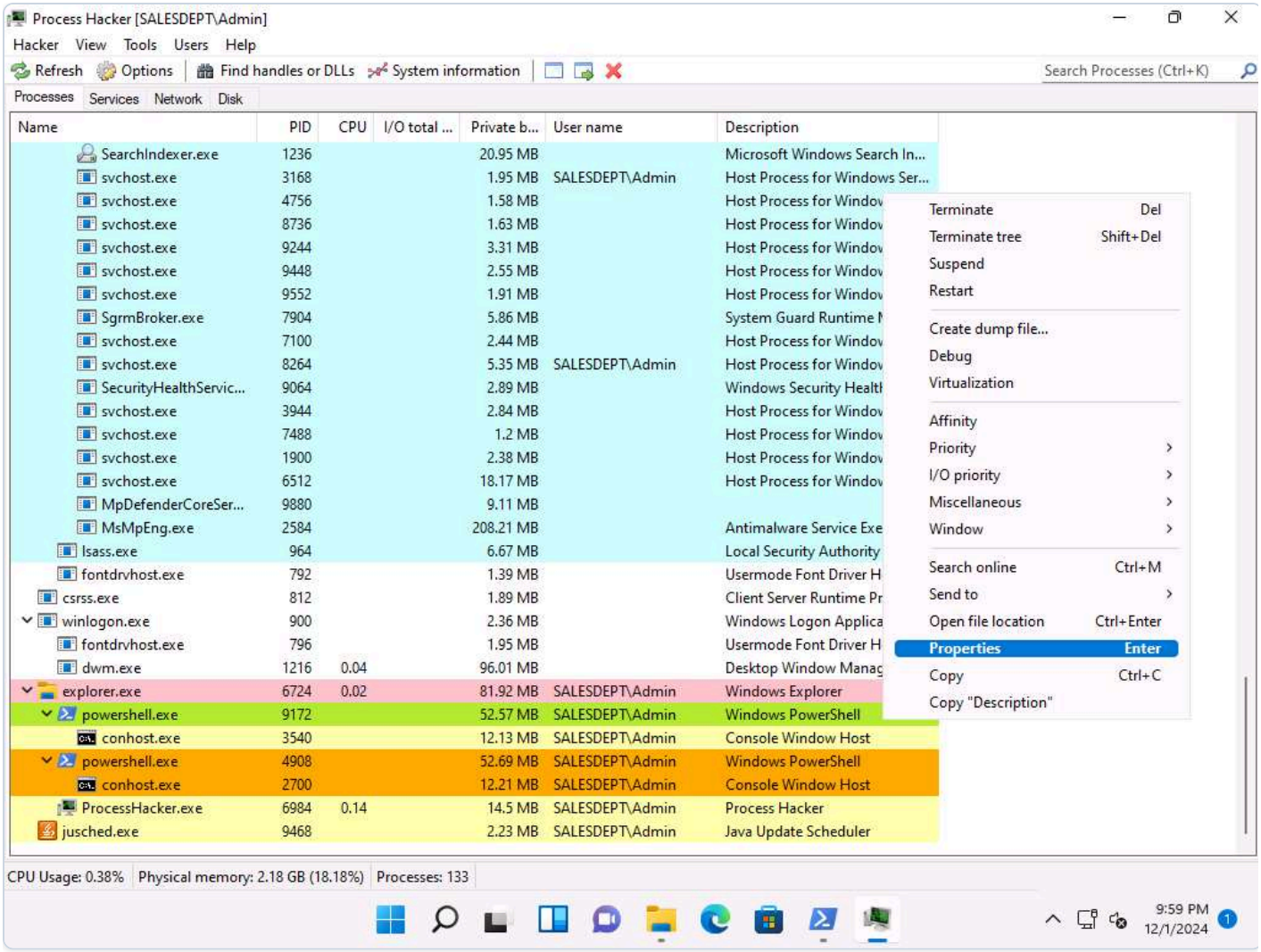
1. Right-click **Windows PowerShell** in search and select **Run as administrator** to open an elevated (High IL) PowerShell window.

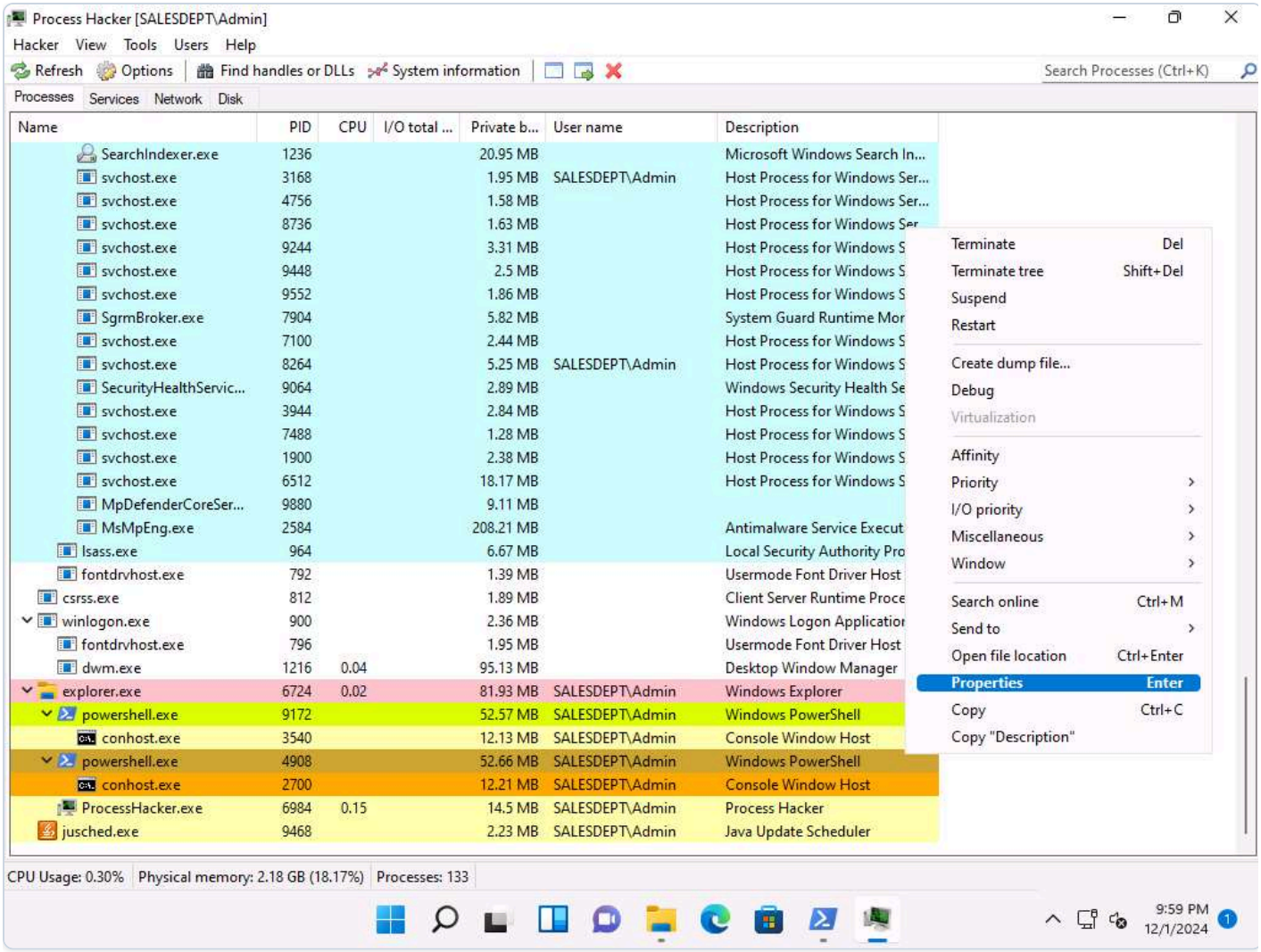


1. Launch **Process Hacker 2**.

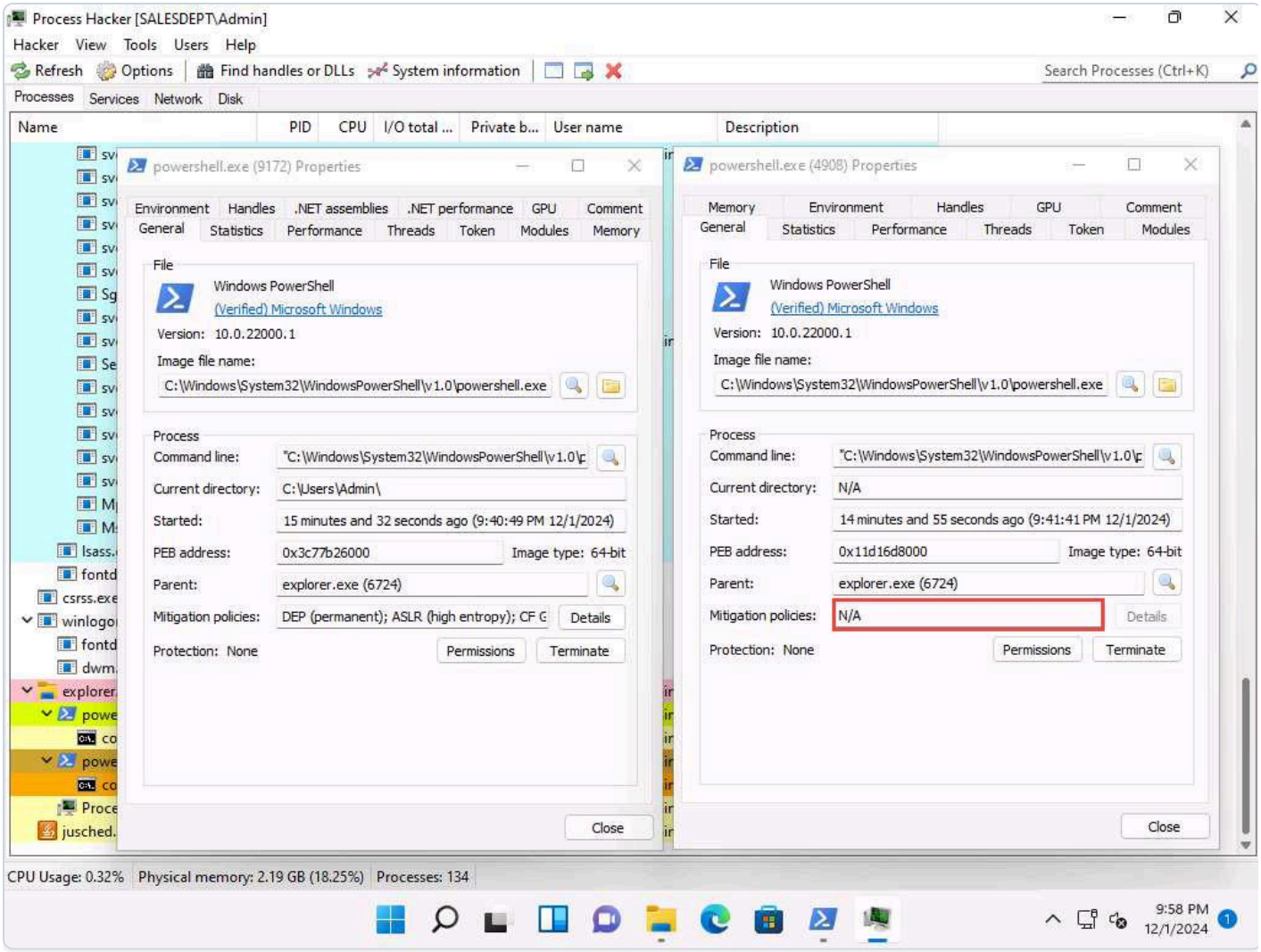


1. Inspect **Properties** for both `powershell.exe` processes.

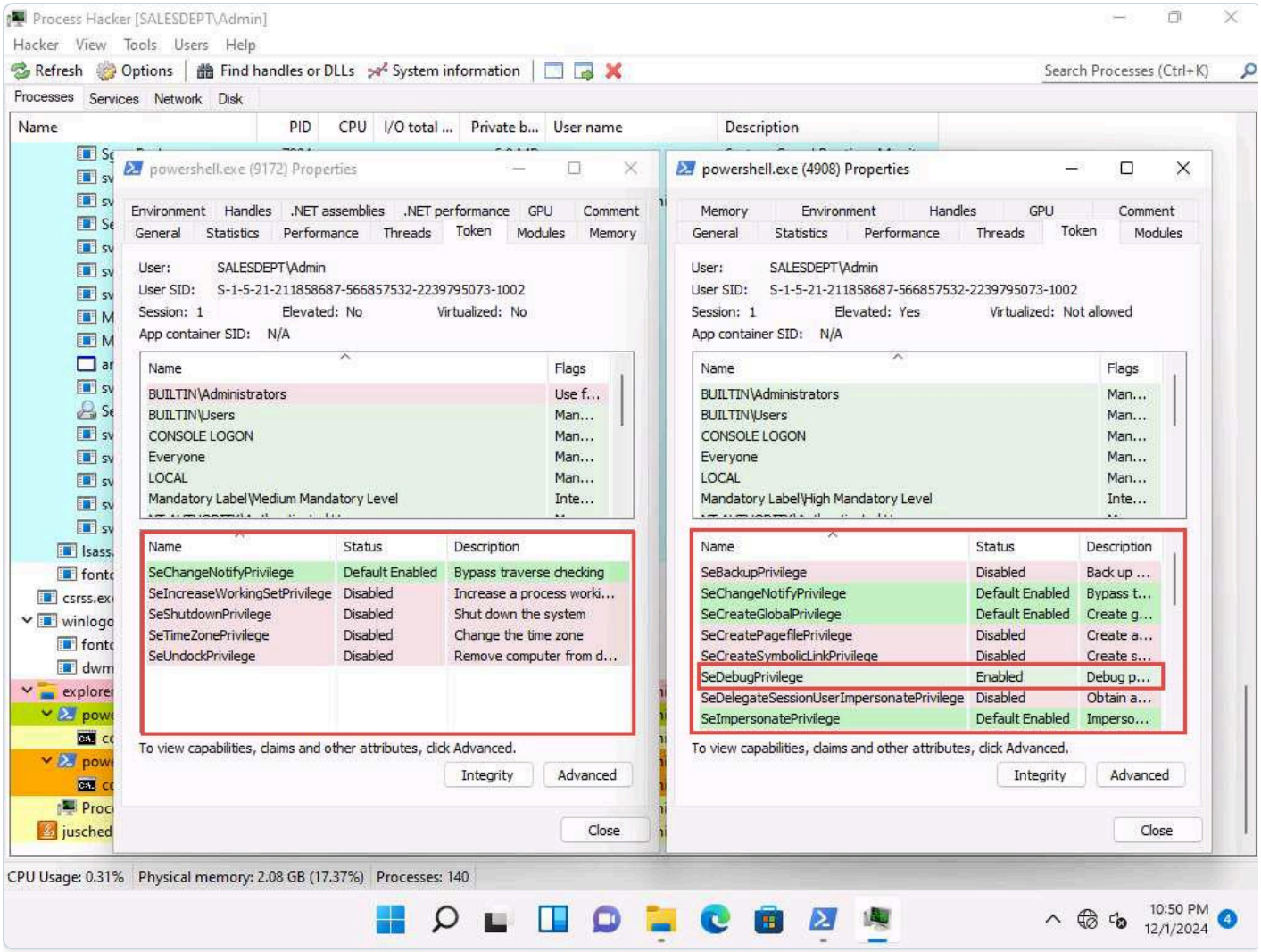

Process Hacker [SALESDEPT\Admin]
Hacker View Tools Users Help
Refresh Options Find handles or DLLs System information
Search Processes (Ctrl+K)
Processes Services Network Disk
Name PID CPU I/O total ... Private b... User name Description
SearchIndexer.exe 1236 20.95 MB Microsoft Windows Search In...
svchost.exe 3168 1.95 MB SALESDEPT\Admin Host Process for Windows Ser...
svchost.exe 4756 1.58 MB Host Process for Windov
svchost.exe 8736 1.63 MB Host Process for Windov
svchost.exe 9244 3.31 MB Host Process for Windov
svchost.exe 9448 2.55 MB Host Process for Windov
svchost.exe 9552 1.91 MB Host Process for Windov
SgrmBroker.exe 7904 5.86 MB System Guard Runtime M
svchost.exe 7100 2.44 MB Host Process for Windov
svchost.exe 8264 5.35 MB SALESDEPT\Admin Host Process for Windov
SecurityHealthServic... 9064 2.89 MB Windows Security Health
svchost.exe 3944 2.84 MB Host Process for Windov
svchost.exe 7488 1.2 MB Host Process for Windov
svchost.exe 1900 2.38 MB Host Process for Windov
svchost.exe 6512 18.17 MB Host Process for Windov
MpDefenderCoreSer... 9880 9.11 MB
MsMpEng.exe 2584 208.21 MB Antimalware Service Exe
lsass.exe 964 6.67 MB Local Security Authority
fontdrvhost.exe 792 1.39 MB Usermode Font Driver H
csrss.exe 812 1.89 MB Client Server Runtime Pr
winlogon.exe 900 2.36 MB Windows Logon Applica
fontdrvhost.exe 796 1.95 MB Usermode Font Driver H
dwm.exe 1216 0.04 96.01 MB Desktop Window Manag
explorer.exe 6724 0.02 81.92 MB SALESDEPT\Admin Windows Explorer
powershell.exe 9172 52.57 MB SALESDEPT\Admin Windows PowerShell
conhost.exe 3540 12.13 MB SALESDEPT\Admin Console Window Host
powershell.exe 4908 52.69 MB SALESDEPT\Admin Windows PowerShell
conhost.exe 2700 12.21 MB SALESDEPT\Admin Console Window Host
ProcessHacker.exe 6984 0.14 14.5 MB SALESDEPT\Admin Process Hacker
jusched.exe 9468 2.23 MB SALESDEPT\Admin Java Update Scheduler
Terminate Del
Terminate tree Shift+Del
Suspend
Restart
Create dump file...
Debug
Virtualization
Affinity
Priority
I/O priority
Miscellaneous
Window
Search online Ctrl+M
Send to
Open file location Ctrl+Enter
Properties Enter
Copy Ctrl+C
Copy "Description"
CPU Usage: 0.38% Physical memory: 2.18 GB (18.18%) Processes: 133
9:59 PM
12/1/2024



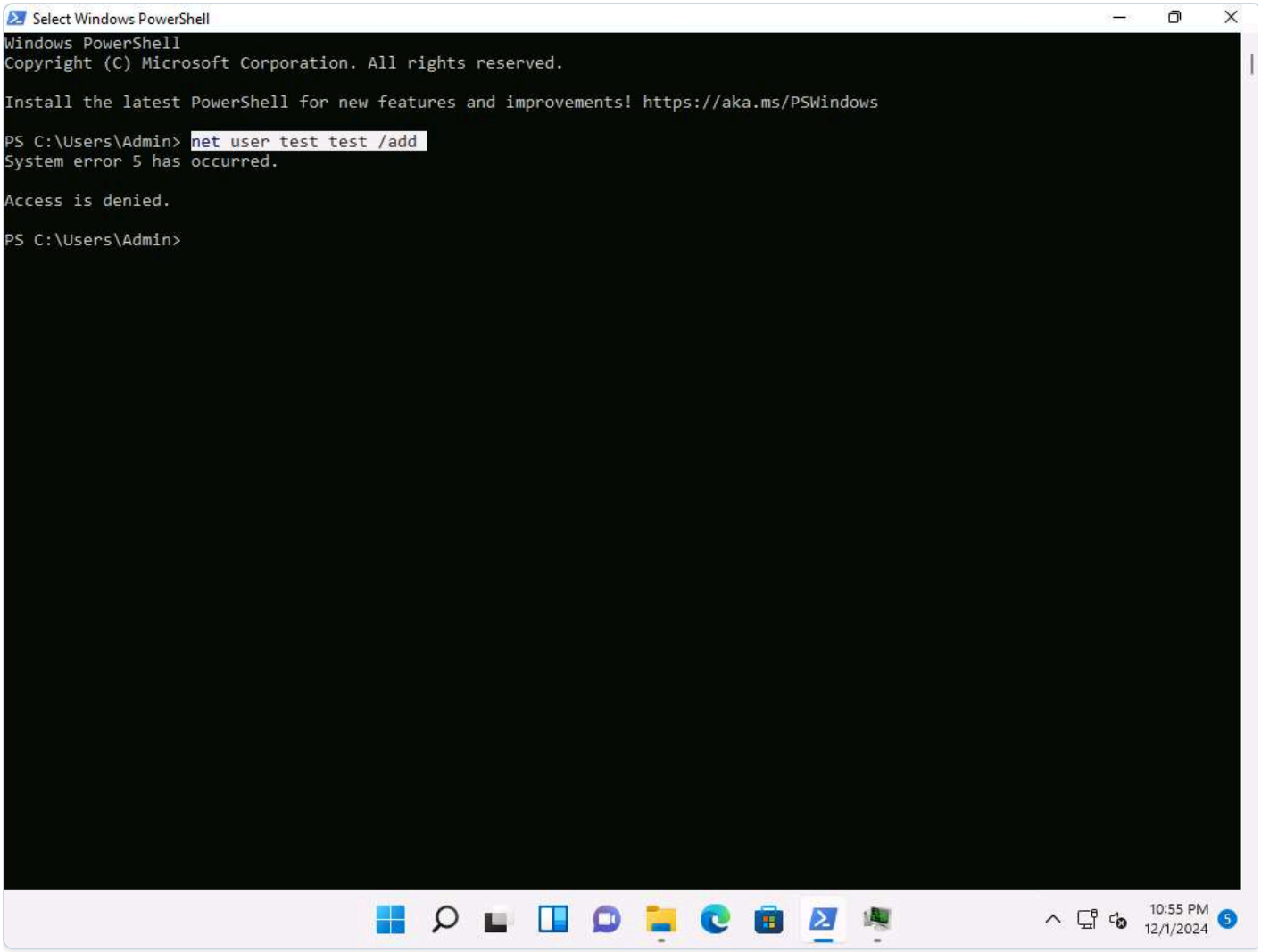
1. Compare mitigation policies between processes. *Observe that for Administrator-level High IL processes, DEP mitigation policies are not enforced (**False**).*



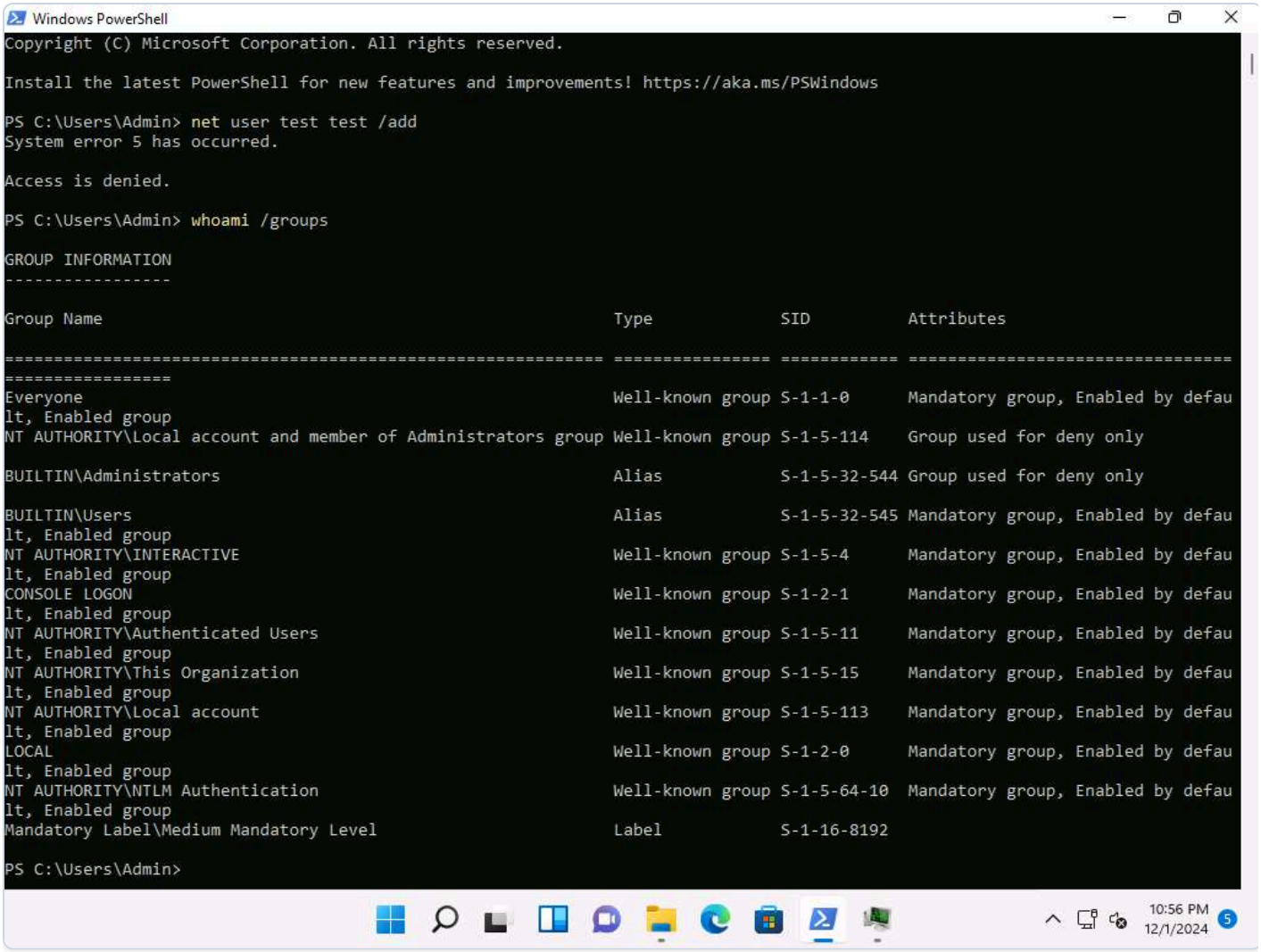
1. Click the **Token** tab for both processes. *Observe that the Elevated Token possesses **SeDebugPrivilege** enabled, whereas the Medium IL Filtered Token lacks administrative privileges.*



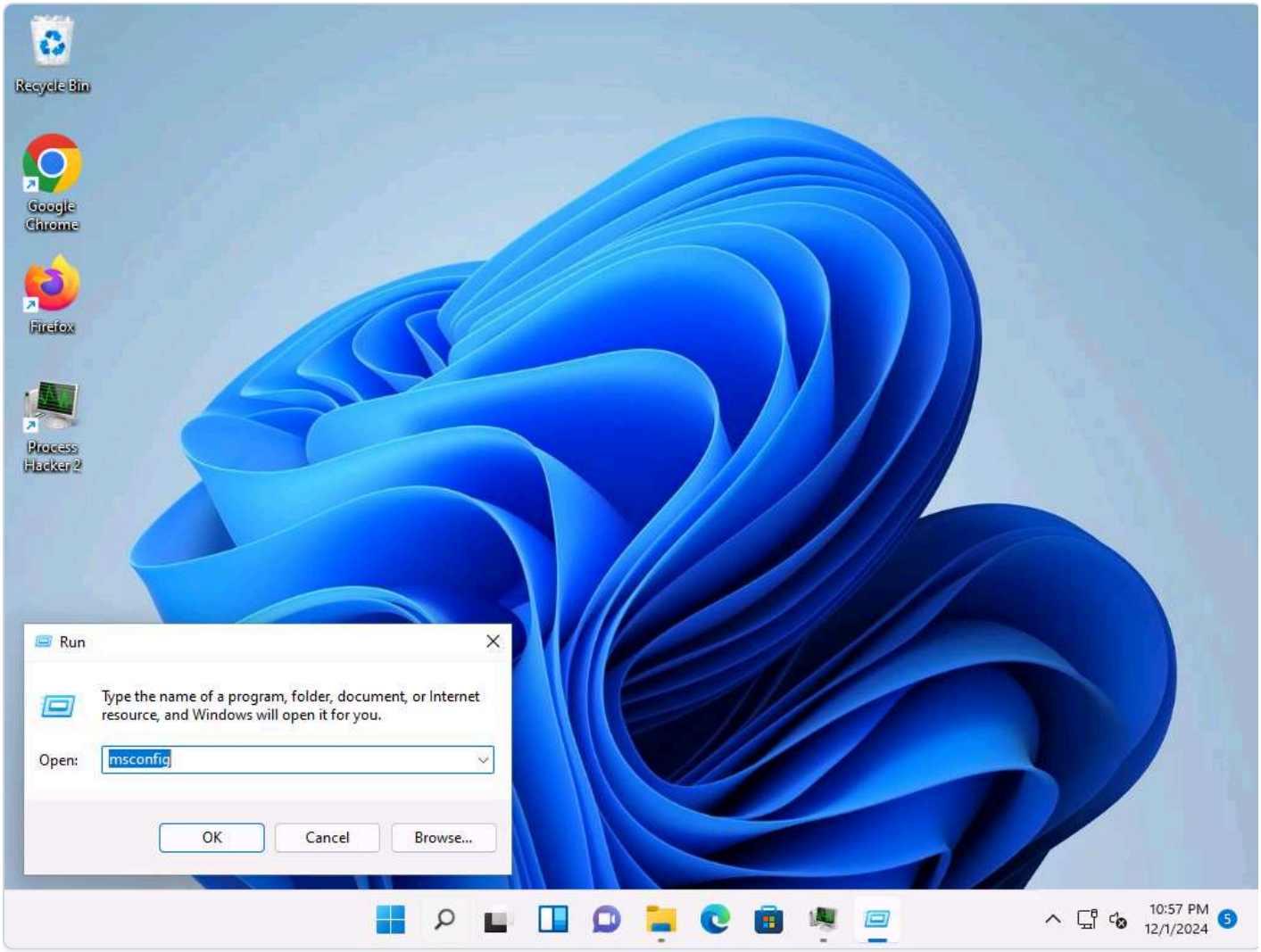
1. In the non-administrator PowerShell window, attempt creating a new user account:
   net user test test /add
   *Access is denied because the current process token runs at **Medium Mandatory Level**.*

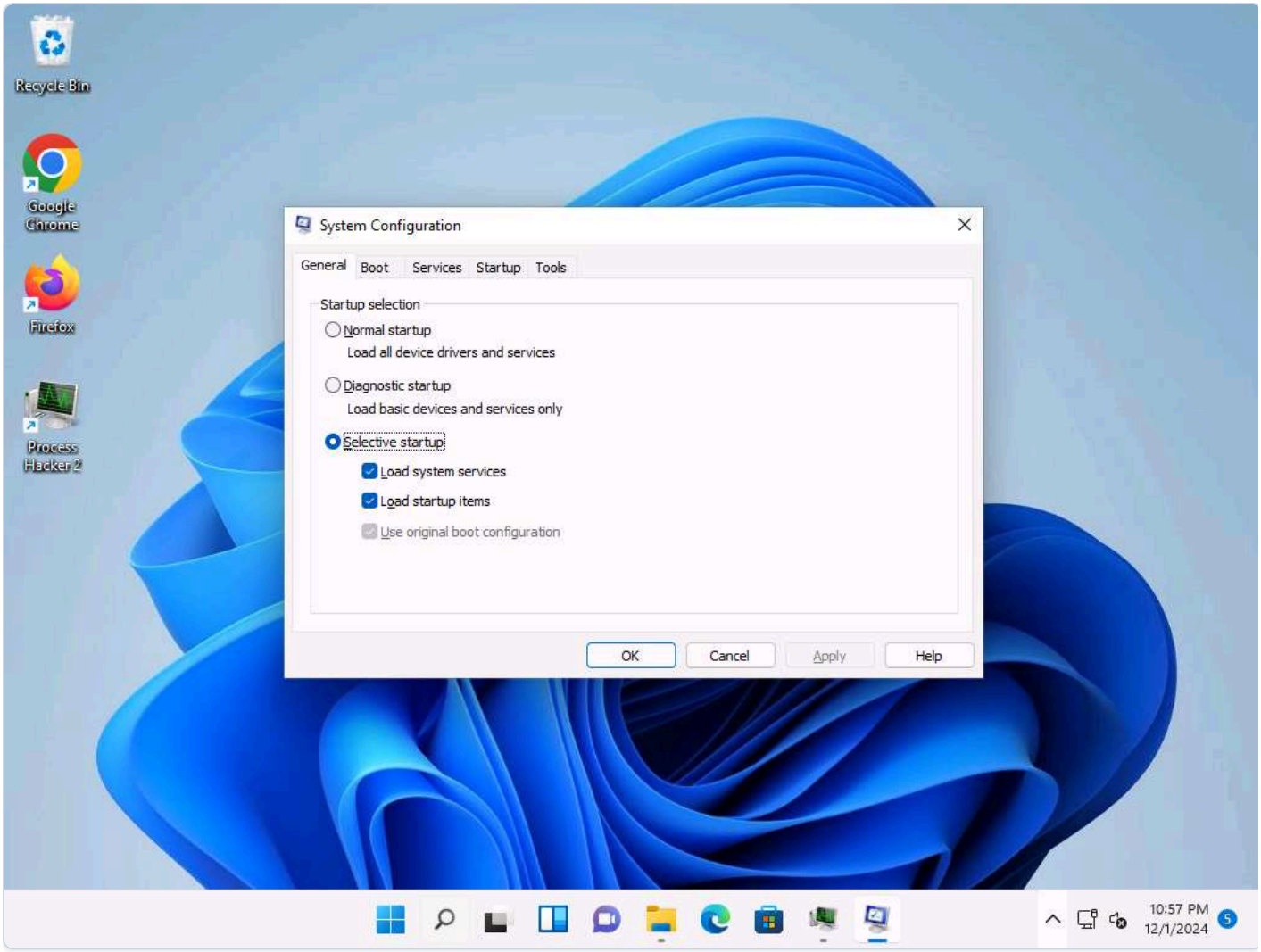


1. Verify integrity level using **whoami /groups**.

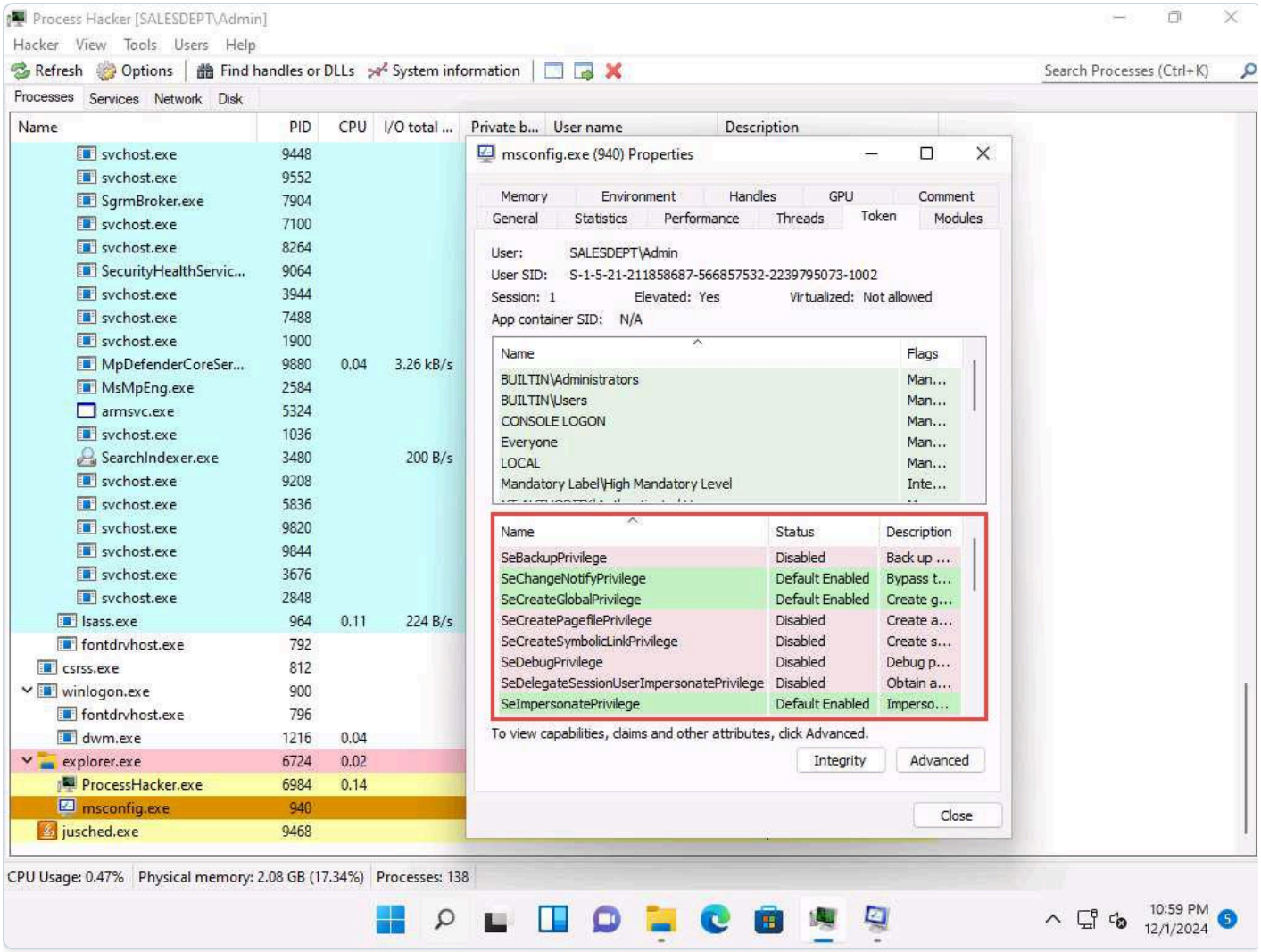


1. Launch `msconfig.exe` via Run dialog (**Win + R $\rightarrow$ msconfig**). *Notice that no UAC prompt appears due to Auto-Elevation.*

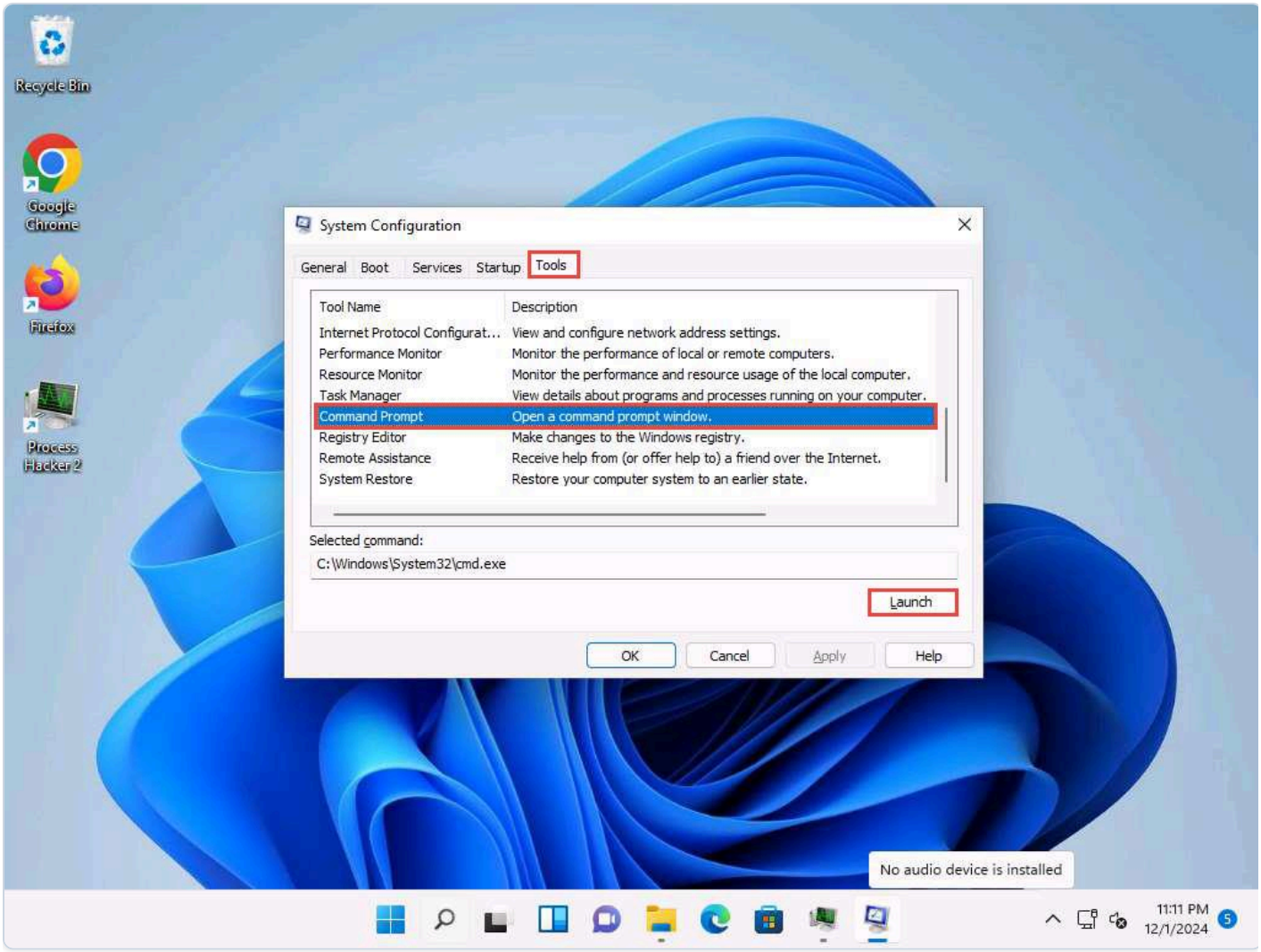

Recycle Bin
Google Chrome
Firefox
Process Hacker 2
Run
Type the name of a program, folder, document, or Internet resource, and Windows will open it for you.
Open:
msconfig
OK
Cancel
Browse...
10:57 PM
12/1/2024

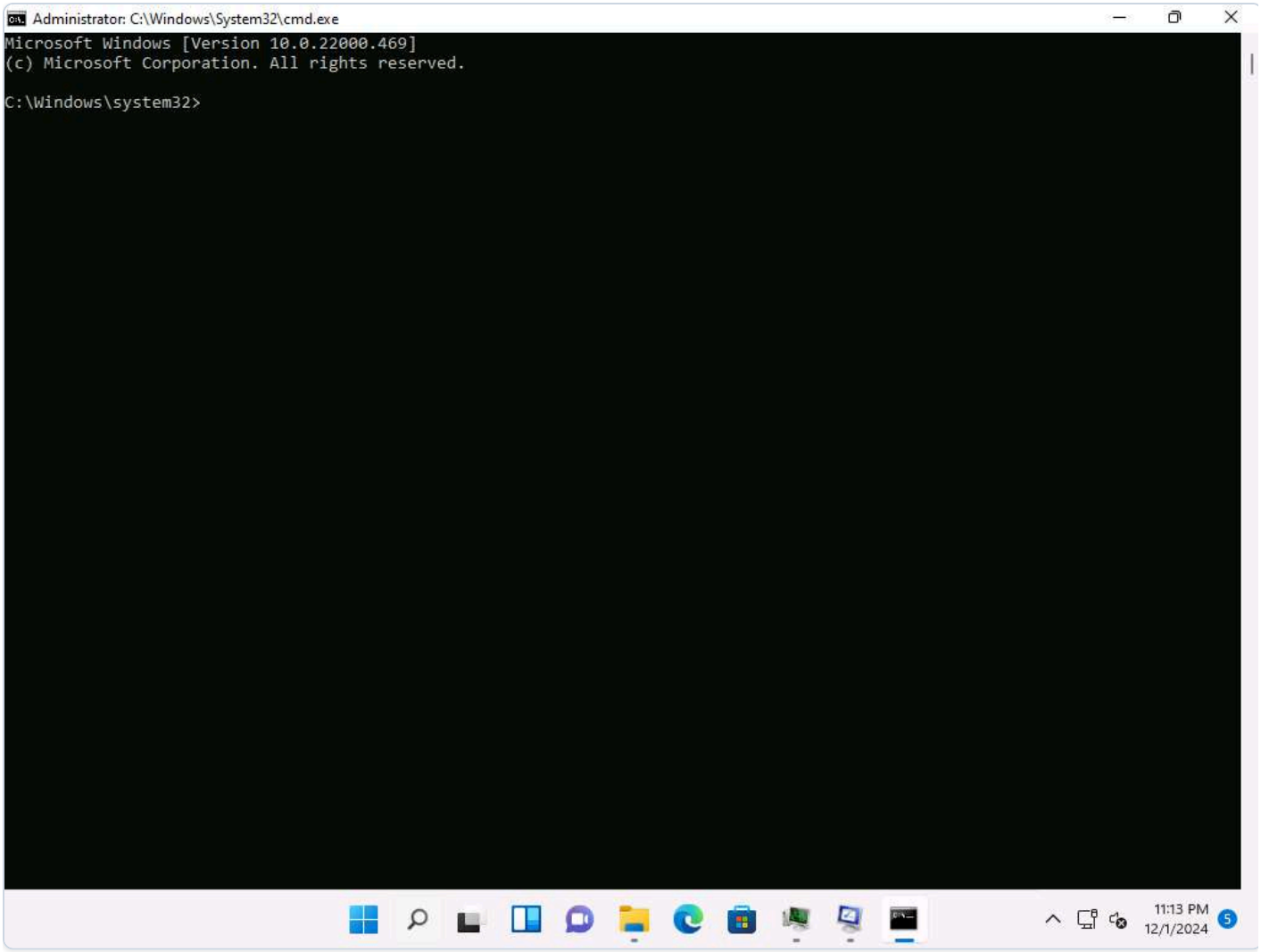


1. Inspect `msconfig.exe` in Process Hacker 2 to verify it runs at **High Integrity Level**.

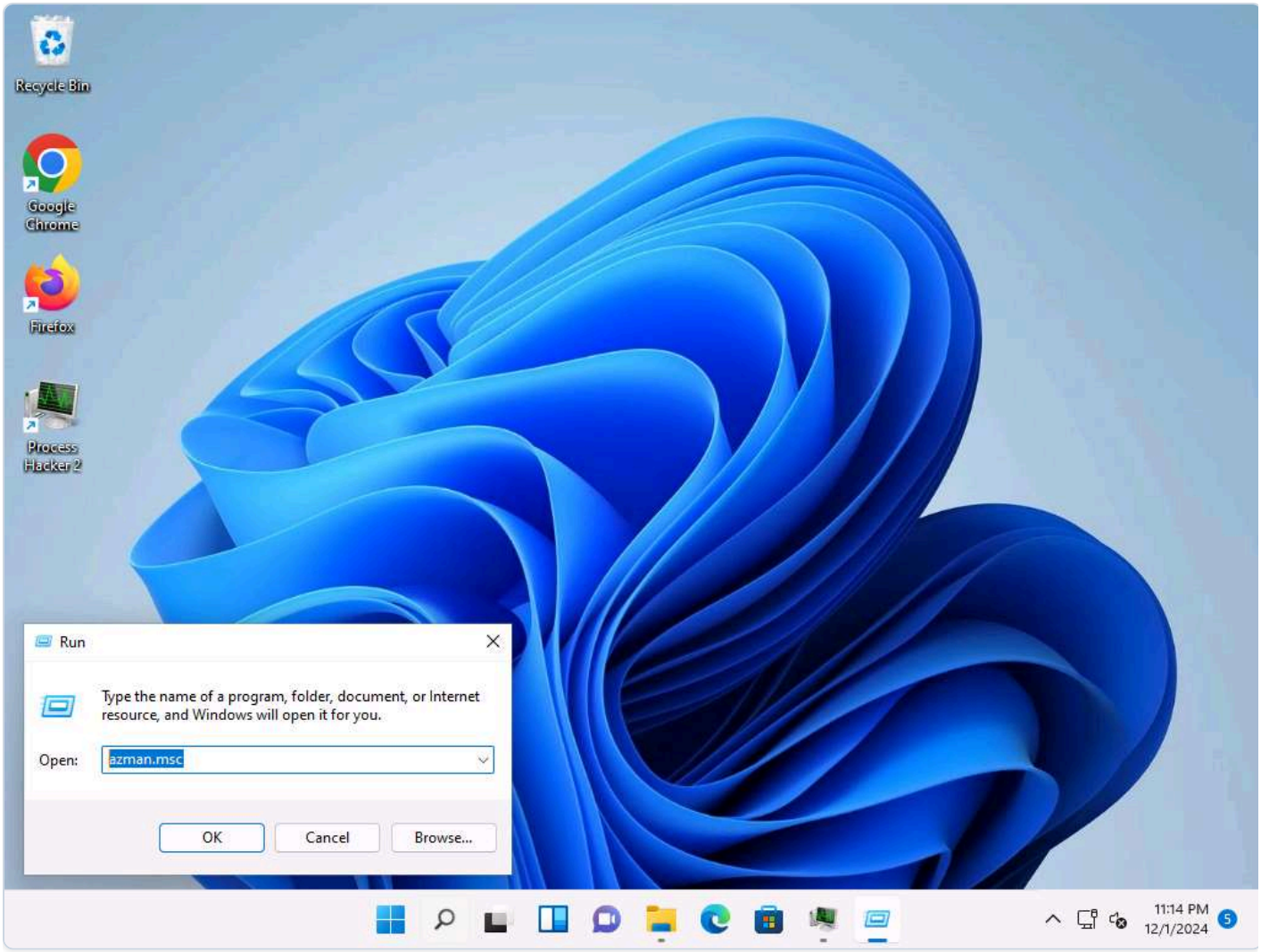


1. Navigate to **Tools** tab in System Configuration window, select **Command Prompt**, and click **Launch**.
   An Administrator Command Prompt opens directly at High Integrity without prompting UAC.

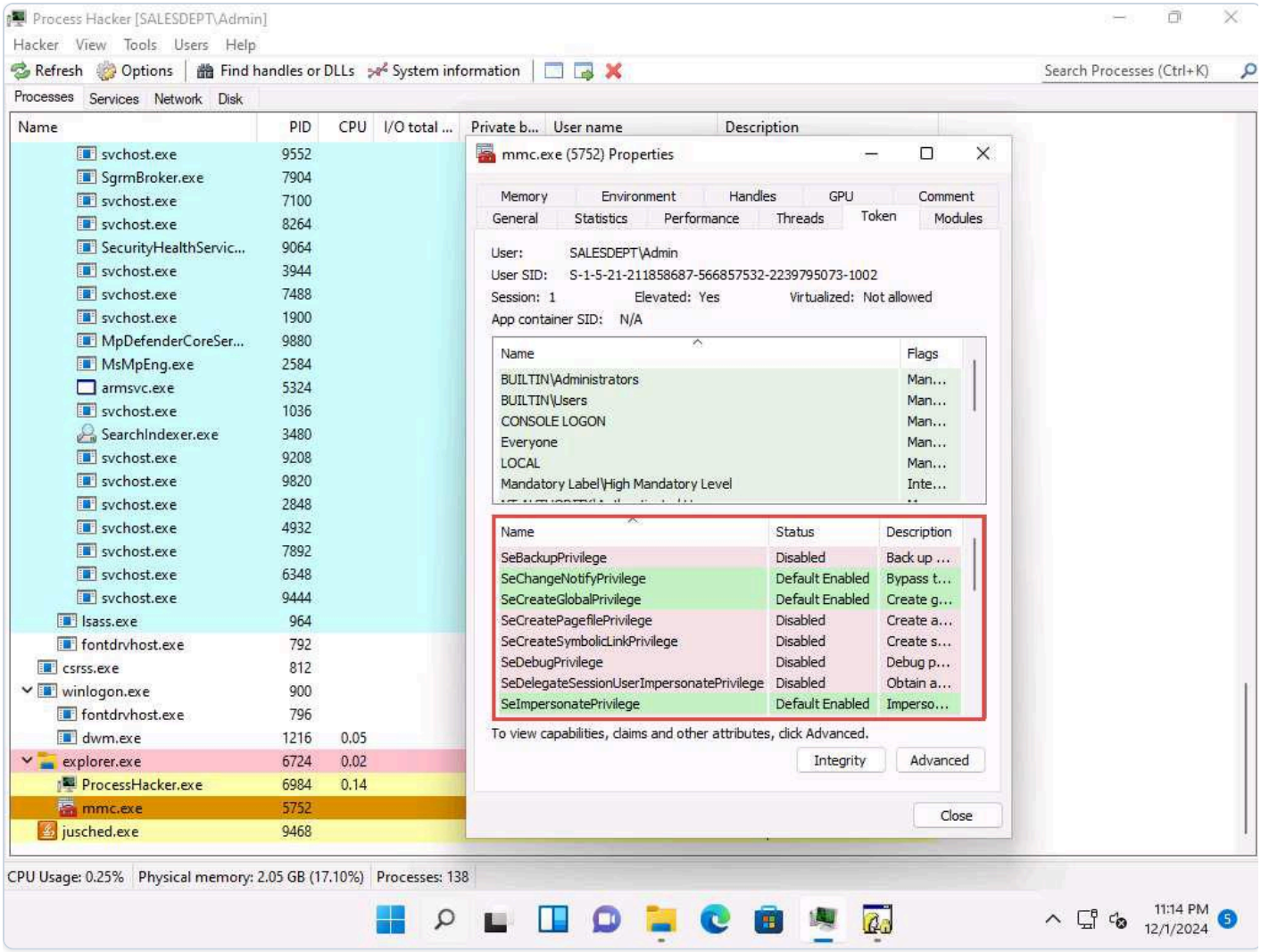

Recycle Bin
Google Chrome
Firefox
Process Hacker 2
System Configuration
General
Boot
Services
Startup
Tools
Tool Name
Description
Internet Protocol Configurat...
View and configure network address settings.
Performance Monitor
Monitor the performance of local or remote computers.
Resource Monitor
Monitor the performance and resource usage of the local computer.
Task Manager
View details about programs and processes running on your computer.
Command Prompt
Open a command prompt window.
Registry Editor
Make changes to the Windows registry.
Remote Assistance
Receive help from (or offer help to) a friend over the Internet.
System Restore
Restore your computer system to an earlier state.
Selected command:
C:\Windows\System32\cmd.exe
Launch
OK
Cancel
Apply
Help
No audio device is installed
11:11 PM
12/1/2024



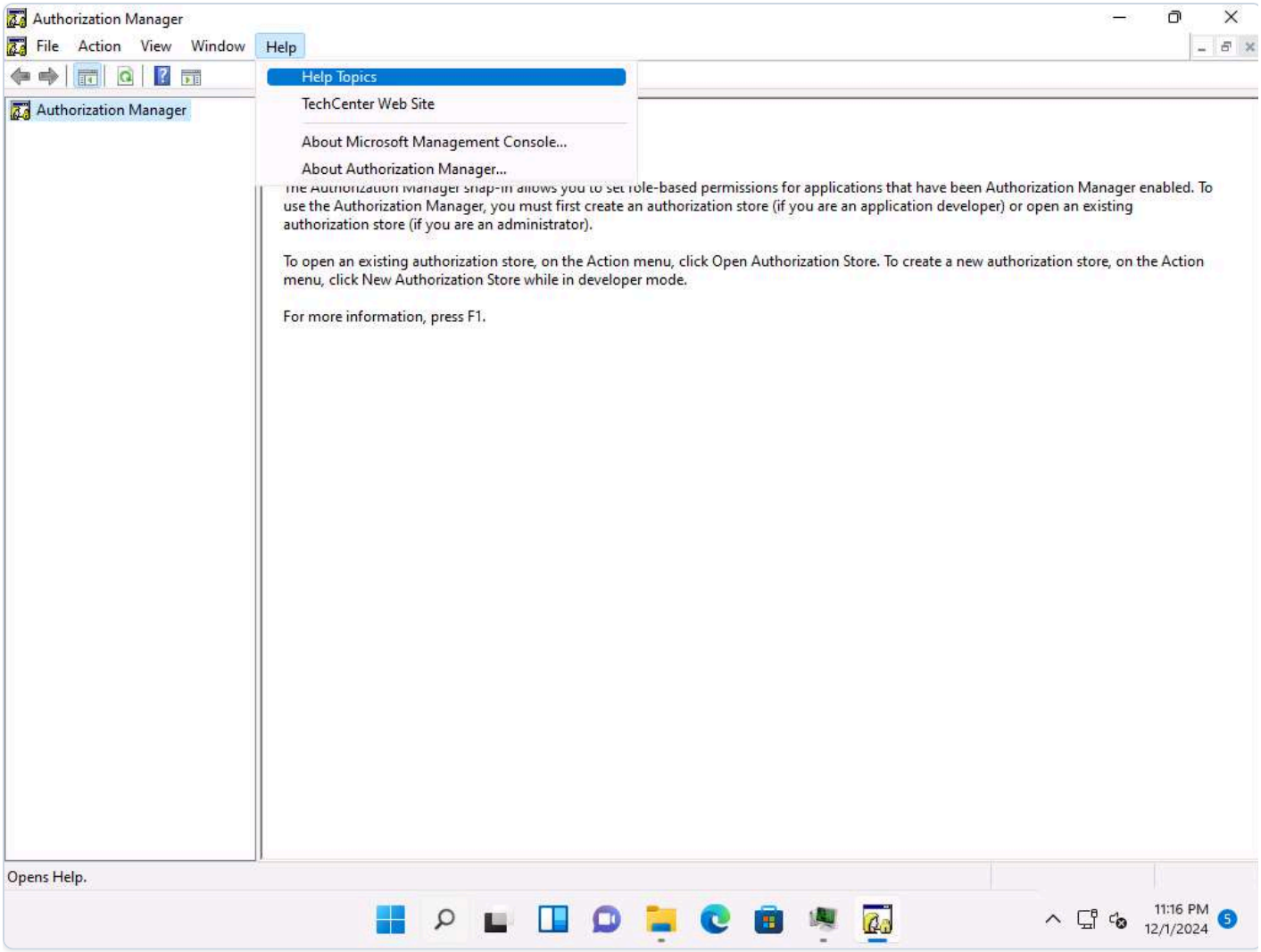
1. Open Authorization Manager via Run dialog (**Win + R $\rightarrow$ azman.msc**).



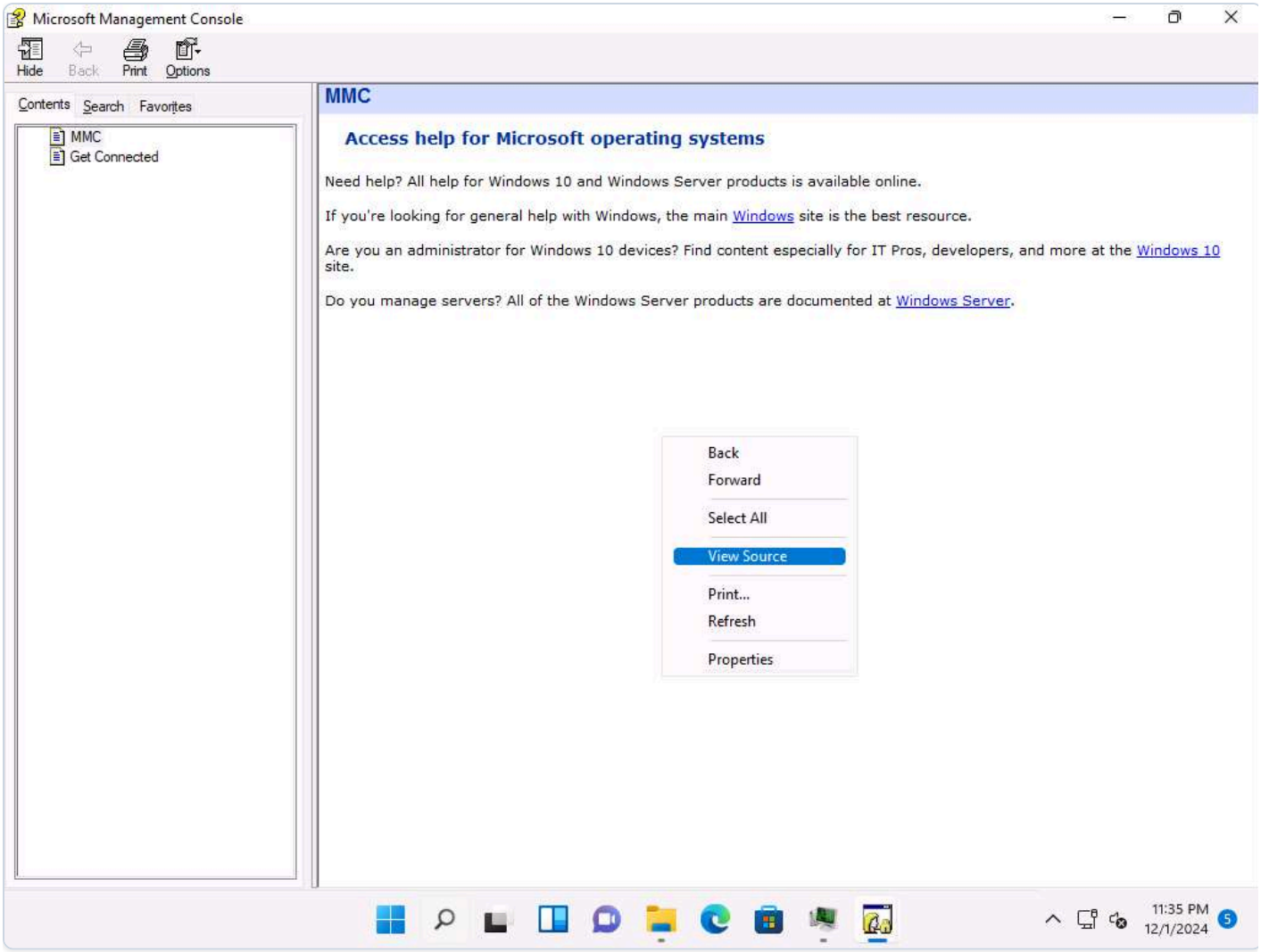
1. Inspect `mmc.exe` in Process Hacker 2 to verify high-integrity token execution.

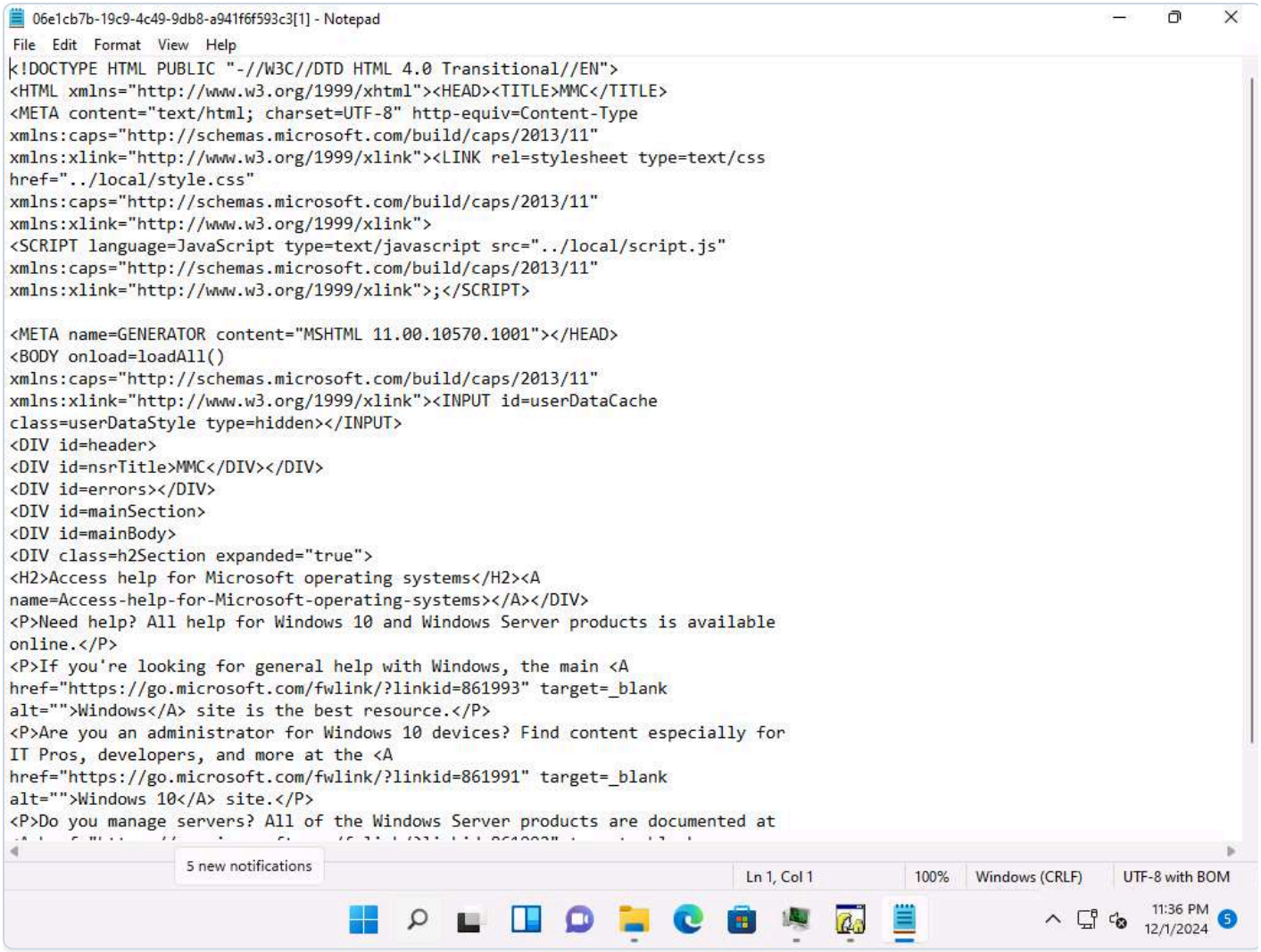


1. Open **Help** $\rightarrow$ **Help Topics** in Authorization Manager window.

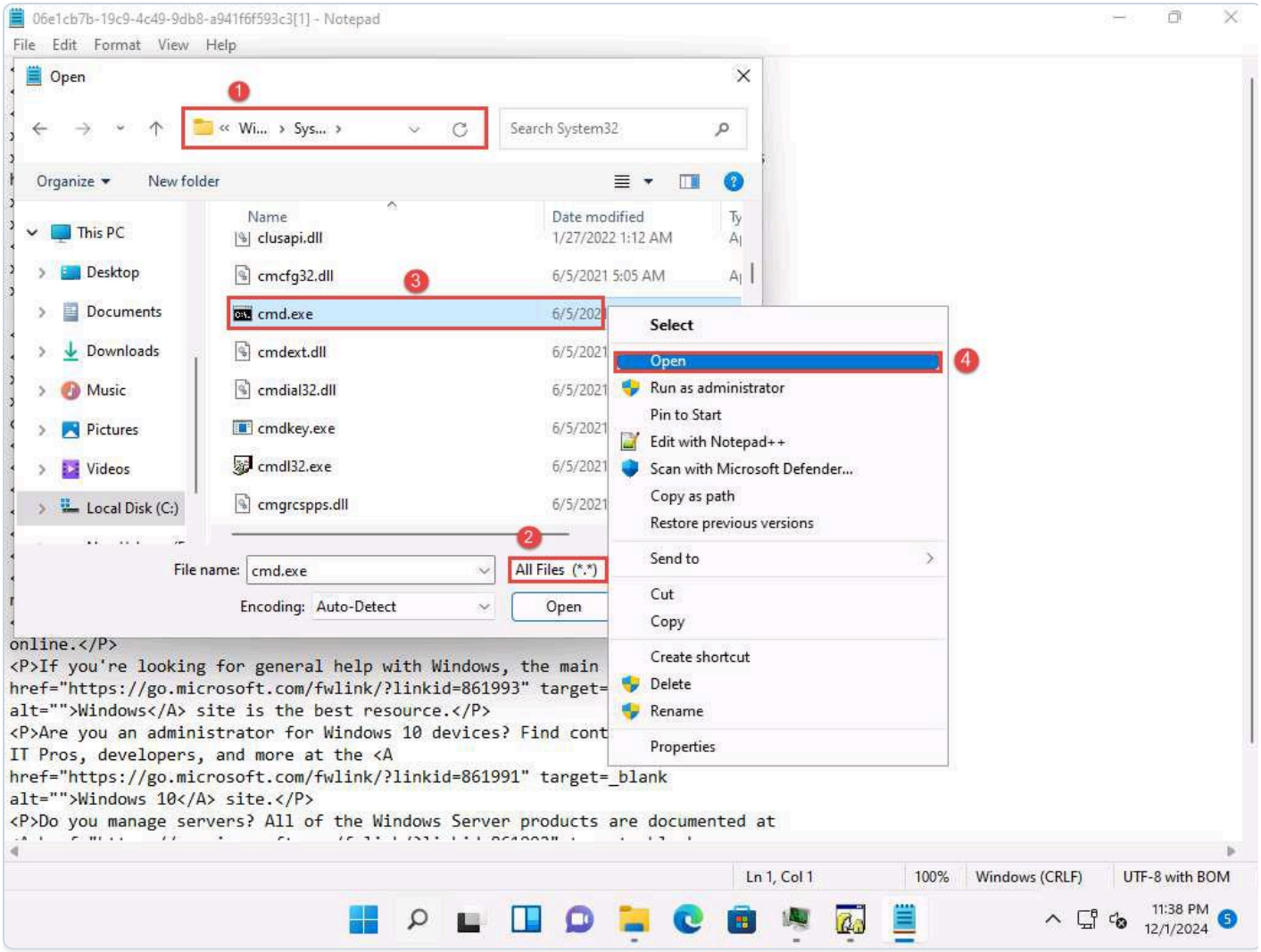


1. Right-click inside the MMC Help window and select **View Source**.
   Notepad opens an XML file with High Integrity inheritance.

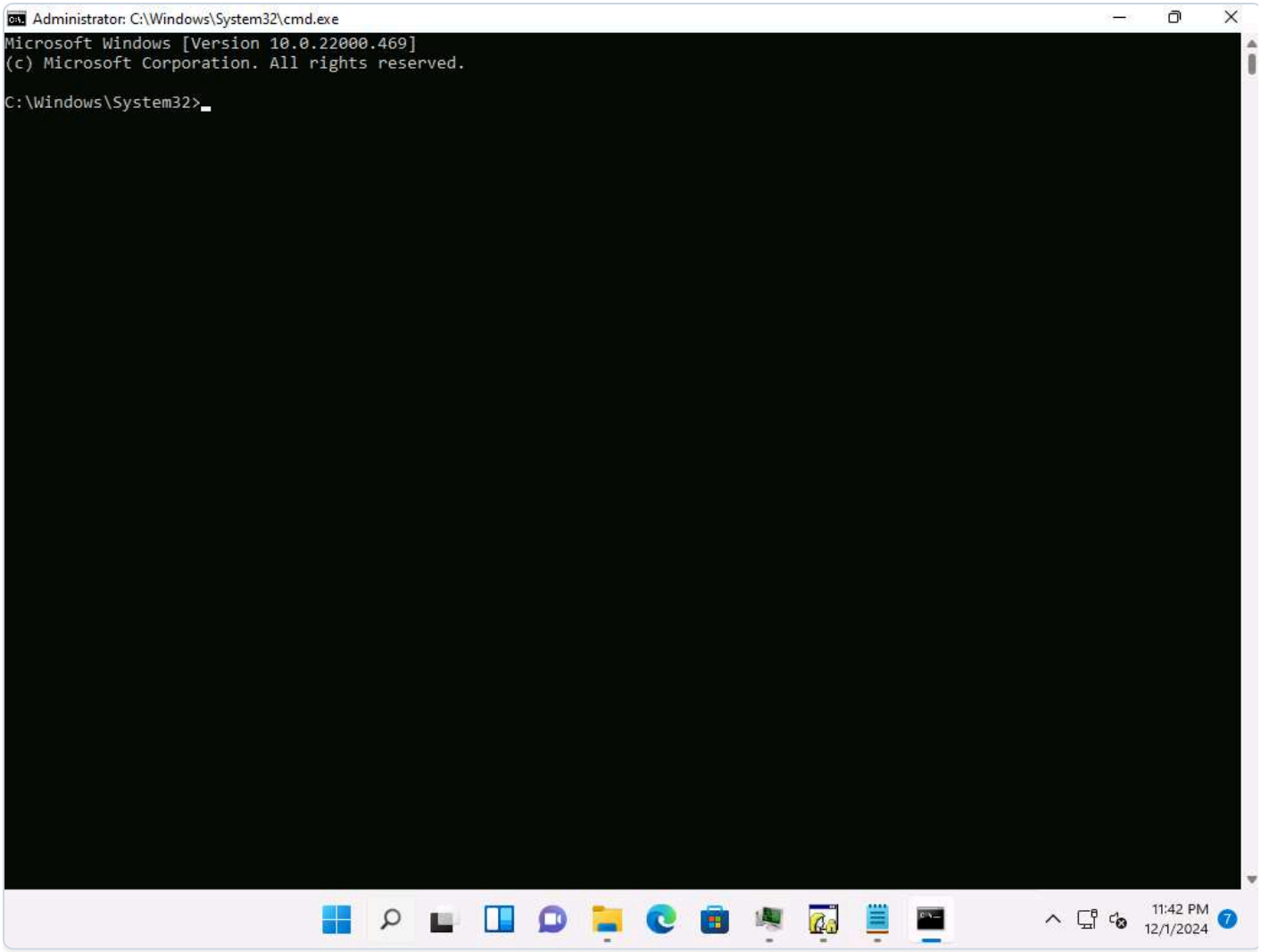

Microsoft Management Console
Hide
Back
Print
Options
Contents
Search
Favorites
MMC
Get Connected
MMC
Access help for Microsoft operating systems
Need help? All help for Windows 10 and Windows Server products is available online.
If you're looking for general help with Windows, the main Windows site is the best resource.
Are you an administrator for Windows 10 devices? Find content especially for IT Pros, developers, and more at the Windows 10 site.
Do you manage servers? All of the Windows Server products are documented at Windows Server.
Back
Forward
Select All
View Source
Print...
Refresh
Properties
11:35 PM
12/1/2024



1. In Notepad, click **File** $\rightarrow$ **Open**, navigate to **C:\Windows\System32**, select **All Files (.)**, locate **cmd.exe**, right-click, and select **Open**.



1. Verify the launched Command Prompt operates at High Integrity Level without UAC prompts.



1. Confirm that DEP enforcement status for Administrator users is **False**. Close all open windows.

This concludes the lab exercise.

> **IMPORTANT:** ❓ **Question 8.8.1.1** On the Sales Department machine, utilize PowerShell, Process Hacker, msconfig, and azman.msc to assess high-integrity level settings in User Account Control (UAC) for potential privilege escalation and elevated access. For Administrator users, determine whether mitigation policies such as Data Execution Prevention (DEP) are enforced (True/False). [!SUCCESS] ✅ **Answer:** **False**